

0.1 Complete List of Motifs

The 54 motifs used in this study span several categories of drought-related regulatory elements: ABA-responsive elements (ABRE, ABRE_core, CE1, CE3, ABRE_like, ACGT_core), dehydration/cold responsive elements (DRE_CRT, DRE_full, LTRE, LTRE_like, CBF_DRE, ICE_box), MYB/MYC binding sites (MYB_core, MYB1AT, MYB2, MYBCORE, MYC), WRKY binding elements (WBOX, WBOX_core), GT elements (GT1, GT1_core), heat shock elements (HSE, HSE_core), ethylene responsive elements (GCC_box, ERE), light/circadian elements (GBOX, IBOX, CAAT_box), stress elements (STRE), wound/pathogen elements (JASE, TBOX), NAC binding sites (NAC_core, NACRS), auxin responsive elements (AuxRE, AuxRE_rev), root-specific motifs (ROOTMOTIF), general elements (AS1, TATA_box, CACTFTPPCA1), and common 4-mer regulatory elements (ACGT, CACG, CATG, CGTG, GCGT, AACA, CAAC, AAAG, GATA, TATA, CAAT, GCCG, CGCG, TGAC, GTAC) [3], [4], [5], [6], [7], [8].

Here we provide the detailed list of the cis-regulatory motifs used in this study and their cross-references to the PLACE [1] and PlantCARE [2] databases. Motif sequences were implemented in the Python pipeline as regular-expression (regex) patterns and scanned on both the forward strand and its reverse complement. The *Motif Key Name* column gives the identifier used in the analysis pipeline, while the *PLACE Key Name* column gives the closest corresponding database entry. PLACE is available at https://rapdb.dna.naro.go.jp/PLACE/place_seq.shtml, and PlantCARE is available at <https://bioinformatics.psb.ugent.be/webtools/plantcare/html/>.

Table 1: ABA-responsive cis-regulatory motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
ABRE	ACGTG[GT][CT]	ABREATCONSENSUS	S000406	PLACE
ABRE_core	ACGTG	ABRELATERD1	S000414	PLACE
CE1	TGCCACCGG	ABRECE1HVA22	S000014	PLACE
CE3	ACGCGTGTC	CE3OSOSEM	S000282	PLACE
ABRE_like	[AC]ACGT[GC][CGT]	ABREZMRAB28_LIKE	S000133	PLACE
ACGT_core	ACGT	ACGTATERD1	S000415	PLACE

Table 2: Dehydration and low-temperature responsive motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
DRE_CRT	CCGAC	LTRECOREATCOR15	S000153	PLACE
DRE_full	[AG]CCGAC	DRECRTCOREAT	S000418	PLACE
LTRE	CCGAC	LTRECOREATCOR15	S000153	PLACE
LTRE_like	ACCGACA	LTREATLTI78	S000157	PLACE
CBF_DRE	GTCGAC	CRTDREHVCBF2	S000411	PLACE

Table 3: MYB transcription factor binding motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
MYB_core	C[ACGT]GTT[AG]	MYBCORE	S000176	PLACE
MYB1AT	[AT]AACCA	MYB1AT	S000408	PLACE
MYB2	TAACTG	MYB2AT	S000177	PLACE
MYBCORE	C[ACGT]GTT[AG]	MYBCORE	S000176	PLACE

Table 4: MYC and E-box related motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
MYC	CA[ACGT][ACGT]TG	MYCCONSENSUSAT	S000407	PLACE
ICE_box	CA[ACGT][ACGT]TG	MYCCONSENSUSAT_ICE1	S000407	PLACE
NACRS	CATGTG	MYCATERD1	S000413	PLACE

Table 5: WRKY and GT transcription factor binding motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
WBOX	TTGAC[CT]	ELRECOREPCRP1	S000142	PLACE
WBOX_core	TGAC	ELRECOREPCRP1_CORE	S000142	PLACE
GT1	G[AG][AT]AA[AT]	GT1CONSENSUS	S000198	PLACE
GT1_core	GAAAAA	GT1GMSCAM4	S000453	PLACE

Table 6: General regulatory and promoter motifs used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
HSE	GAA[ACGT][ACGT]TTC	HSE	S000054	PLACE
HSE_core	[ACGT]GAA[ACGT]	HSE_CORE	S000054	PLACE
GCC_box	GCCGCC	GCCCORE	S000430	PLACE
ERE	A[AT]TTCAAA	ERELEE4	S000037	PLACE
GBOX	CACGTG	CACGTGMOTIF	S000042	PLACE
IBOX	GATAAG	IBOX	S000124	PLACE
CAAT_box	CCAAT	CCAATBOX1	S000030	PLACE
TATA_box	TATAAA	TATABOX2	S000132	PLACE
STRE	AGGGG	STRE	PlantCARE	PlantCARE

Table 7: Additional regulatory motifs including NAC, auxin, and root-related elements.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
JASE	CGTCA	JASE1ATOPR1	S000388	PLACE
TBOX	ACTTTG	TBOX	PlantCARE	PlantCARE
NAC_core	CACG	NACCORE_CACG	S000042	PLACE
AuxRE	TGTCTC	ARFAT	S000270	PLACE
AuxRE_rev	GAGACA	ARFAT_REVCOMP	PlantCARE	PlantCARE
ROOTMOTIF	ATATT	AT1BOX_CORE	S000025	PLACE
AS1	TGACG	AUXRETGA1GMGH3_CORE	S000234	PLACE
CACTFTPPCA1	[CT]ACT	CACTFTPPCA1	S000449	PLACE

Table 8: Simple 4-mer regulatory sequence features used in this study.

Motif Key Name	Regex / Sequence	PLACE Key Name	PLACE ID	Source
ACGT	ACGT	ACGTATERD1_4MER	S000415	PLACE
CACG	CACG	CACGTGMOTIF_CORE	S000042	PLACE
CATG	CATG	MYCATERD1_CORE	S000413	PLACE
CGTG	CGTG	ABRE_CGTG_CORE	S000406	PLACE
GCGT	GCGT	CE3_GCGT_CORE	S000282	PLACE
AACA	AACA	MYBGAHV_CORE	S000181	PLACE
CAAC	CAAC	CAAC_PLANTCARE	PlantCARE	PlantCARE
AAAG	AAAG	DOFCOREZM	S000265	PLACE
GATA	GATA	GATABOX	S000039	PLACE
TATA	TATA	TATABOX2_CORE	S000132	PLACE
CAAT	CAAT	CCAATBOX1_CORE	S000030	PLACE
GCCG	GCCG	AGCBOXNPGLB_CORE	S000232	PLACE
CGCG	CGCG	CGCGBOXAT_CORE	S000501	PLACE
TGAC	TGAC	ELRECOREPCR1_CORE	S000142	PLACE
GTAC	GTAC	CURECORECR	S000493	PLACE

Bibliography

- [1] Higo, K., Ugawa, Y., Iwamoto, M., and Higo, H. (1998). “PLACE: a database of plant cis-acting regulatory DNA elements.” *Nucleic Acids Research*, 26(1), 358–359.
- [2] Rombauts, S., Déhais, P., Van Montagu, M., and Rouzé, P. (1999). “PlantCARE, a plant cis-acting regulatory element database.” *Nucleic Acids Research*, 27(1), 295–296.
- [3] Mundy, J., Yamaguchi-Shinozaki, K., and Chua, N.H. (1990). “Nuclear proteins bind conserved elements in the abscisic acid-responsive promoter of a rice *rab* gene.” *Proceedings of the National Academy of Sciences USA*, 87(4), 1406–1410.
- [4] Yamaguchi-Shinozaki, K. and Shinozaki, K. (1994). “A novel *cis*-acting element in an *Arabidopsis* gene is involved in responsiveness to drought, low-temperature, or high-salt stress.” *The Plant Cell*, 6(2), 251–264.
- [5] Baker, S.S., Wilhelm, K.S., and Thomashow, M.F. (1994). “The 5′-region of *Arabidopsis thaliana cor15a* has *cis*-acting elements that confer cold-, drought- and ABA-regulated gene expression.” *Plant Molecular Biology*, 24(5), 701–713.
- [6] Abe, H., Yamaguchi-Shinozaki, K., Urao, T., Iwasaki, T., Hosokawa, D., and Shinozaki, K. (1997). “Role of *Arabidopsis* MYC and MYB homologs in drought- and abscisic acid-regulated gene expression.” *The Plant Cell*, 9(10), 1859–1868.
- [7] Abe, H., Urao, T., Ito, T., Seki, M., Shinozaki, K., and Yamaguchi-Shinozaki, K. (2003). “*Arabidopsis* AtMYC2 (bHLH) and AtMYB2 (MYB) function as transcriptional activators in abscisic acid signaling.” *The Plant Cell*, 15(1), 63–78.
- [8] Shinozaki, K., Yamaguchi-Shinozaki, K., and Seki, M. (2003). “Regulatory network of gene expression in the drought and cold stress responses.” *Current Opinion in Plant Biology*, 6(5), 410–417.